ALEJANDRO SERRANO RUIBAL

Software Engineering (UCM) · Business & Technology (UDIMA)

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PROFILE

Software engineering student with a business background, motivated by building technology that solves real problems for people and organisations. I take ownership of my work and see it through until it delivers results. I work well in teams: I listen, coordinate with others and make sure my part fits with theirs. I start from the real problem and the people who will use the product, not from the technology. I communicate clearly with technical and non-technical people, adapt quickly to new tools and environments, and stay calm and organised under pressure. Used to international environments, in Spanish and English.

AWARDS & RECOGNITION

- **Winning project** — Escuela de la Energía, Fundación EDP & Ashoka (AlGas)
- **Best Pitch Award and 2nd place overall** — Iberoamérica 2030 Hackathon, Club Hispanidad Futura (Embarca)
- **Scholarship** — HackMeridian Lisbon 2026, Stellar ecosystem hackathon (Phonendo)
- **Honor Roll Certificate** — Academica International Studies, dual U.S. diploma

SELECTED PROJECTS

Phonendo · *Private health data, verifiable payments on Stellar* — Project lead · HackMeridian Lisbon

- Led the project from idea to working demo in one weekend: a sensor proves that a patient completed a cardiac rehab session without revealing their health data, and a smart contract verifies the proof and pays them in USDC.
- Designed the Groth16 zero-knowledge circuit (Circom, ~2K constraints, BLS12-381, Poseidon commitment) and wrote the Soroban contract in Rust that verifies proofs on-chain and blocks double claims. Made the first on-chain ZK claim on Stellar testnet.

AlGas · *Local electricity from sargassum* — Co-lead · Fundación EDP & Ashoka

- Co-leading a project that turns the sargassum flooding Caribbean beaches into electricity for coastal communities, using small gasification units on the coast, starting in Guayacanes and Juan Dolio (Dominican Republic).
- Built the problem analysis, the solution design with every figure backed by official sources, and the brand. Selected and funded by Fundación EDP and Ashoka to start its validation phase.

Mapping Blockchain Ecosystem · *Final-year research project* — Python · Pandas · NetworkX

- Building reproducible pipelines and graph models over governance data from 85K+ DAOs, 370K+ proposals and millions of votes, defining measurable relationships between organizations and validating them before drawing conclusions.

La Abuelita · *Multi-agent negotiation system* — Claude Hackathon Madrid · 48 h

- A multi-agent system in Python where autonomous buyer and seller agents negotiate, analyse market activity and share what they learn, adjusting each decision to market data and previous negotiations. The key finding was that better information, not faster responses, was what limited the agents. Built in a team of three.

Embarca · *Real company challenges for young talent* — Iberoamérica 2030 Hackathon

- Built the feature where a company describes its need in its own words and AI (Claude API) turns it into a ready-to-publish challenge with brief, deliverable, deadline and rubric (Next.js, TypeScript, Supabase). Also worked on the 90-day pilot plan for the pitch.

Quadratic Voting DAO · *On-chain governance* — Solidity · ERC-20

- Implemented quadratic voting in Solidity smart contracts with ERC-20 tokens, from design to a complete implementation.

EDUCATION

BSc in Software Engineering — Complutense University of Madrid — Sep 2023 – Jul 2027 (expected)
Most coursework in English: algorithms and data structures, databases, software architecture, blockchain.

BSc in Business & Technology — UDIMA — Sep 2024 – Present
Studied simultaneously with Software Engineering: management, digital innovation and technology in organizations.

Technical High School — CEU Claudio Coello, Madrid — Sep 2021 – Jun 2023

Dual U.S. High School Diploma — Academica International Studies — Sep 2019 – Jul 2022
Taken entirely in English alongside the Spanish system.

EXPERIENCE

Robotics & Programming Teacher — Camp Tecnológico, Madrid — Sep 2022 – Jun 2023
Taught robotics and programming to primary and secondary students, in Spanish and in English.

SKILLS

Programming: C++, Java, Python, Rust, C, TypeScript / JavaScript, SQL, Solidity
Blockchain & ZK: Stellar, Soroban, Circom, snarkjs, Ethereum, ERC-20, smart contracts
Web & AI: Next.js, React, Tailwind CSS, Supabase, Claude API, multi-agent LLM systems
Data & cloud: Pandas, NetworkX, Jupyter, MongoDB, Docker, AWS (ECR, ECS), Git / GitHub, Scrum
Languages: Spanish (native) · English (C1, seven years of education in English)